Mechatronics Design Laboratory I

ENME 561/661

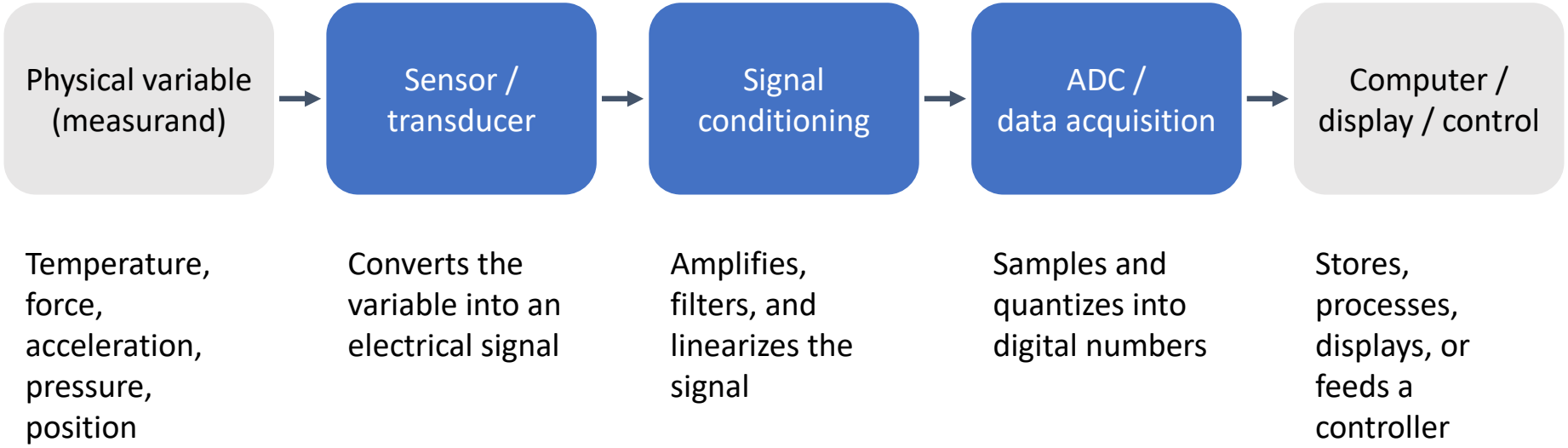
Mahdis Bisheban
Associate Professor
Department of Mechanical and Manufacturing Engineering

Mahdis.bisheban@ucalgary.ca

2026/11/06

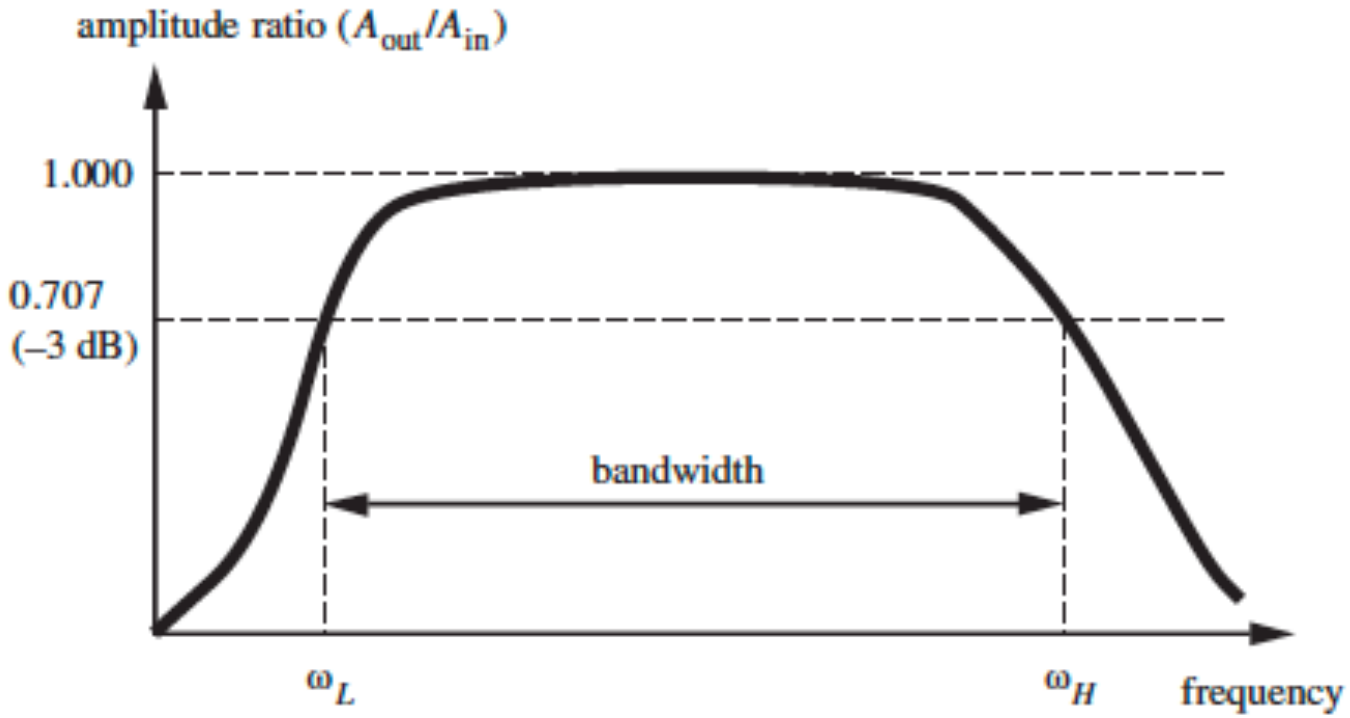
What is a Measurement System?

A measurement system converts a physical quantity (the measurand) into a signal that a person or a computer can read, record, or use for control.

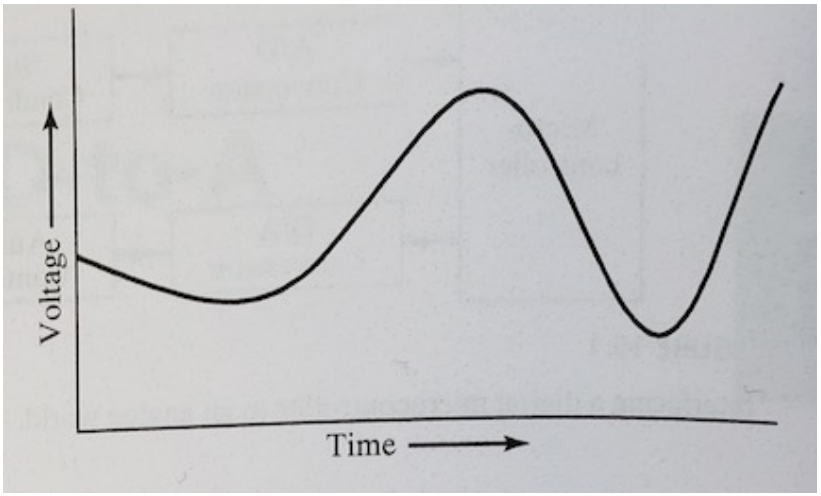


Bandwidth and Frequency Response of a measurement system

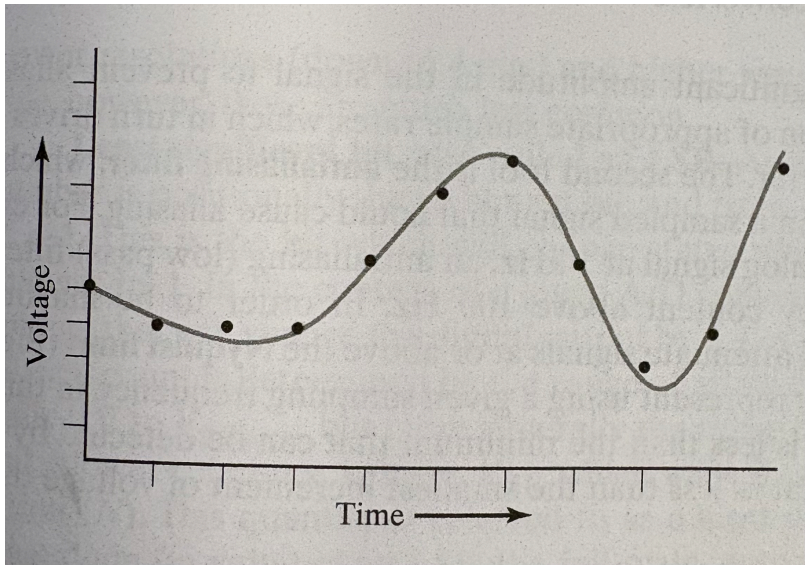
- Ideally, a measurement system should replicate all frequency components of a signal.
- Real systems, however, have limitations in their ability to reproduce all frequencies.
- Bandwidth: the range of frequencies a system can adequately reproduce (not attenuated more than 3db)



Analog signal
(continuous in time and amplitude)

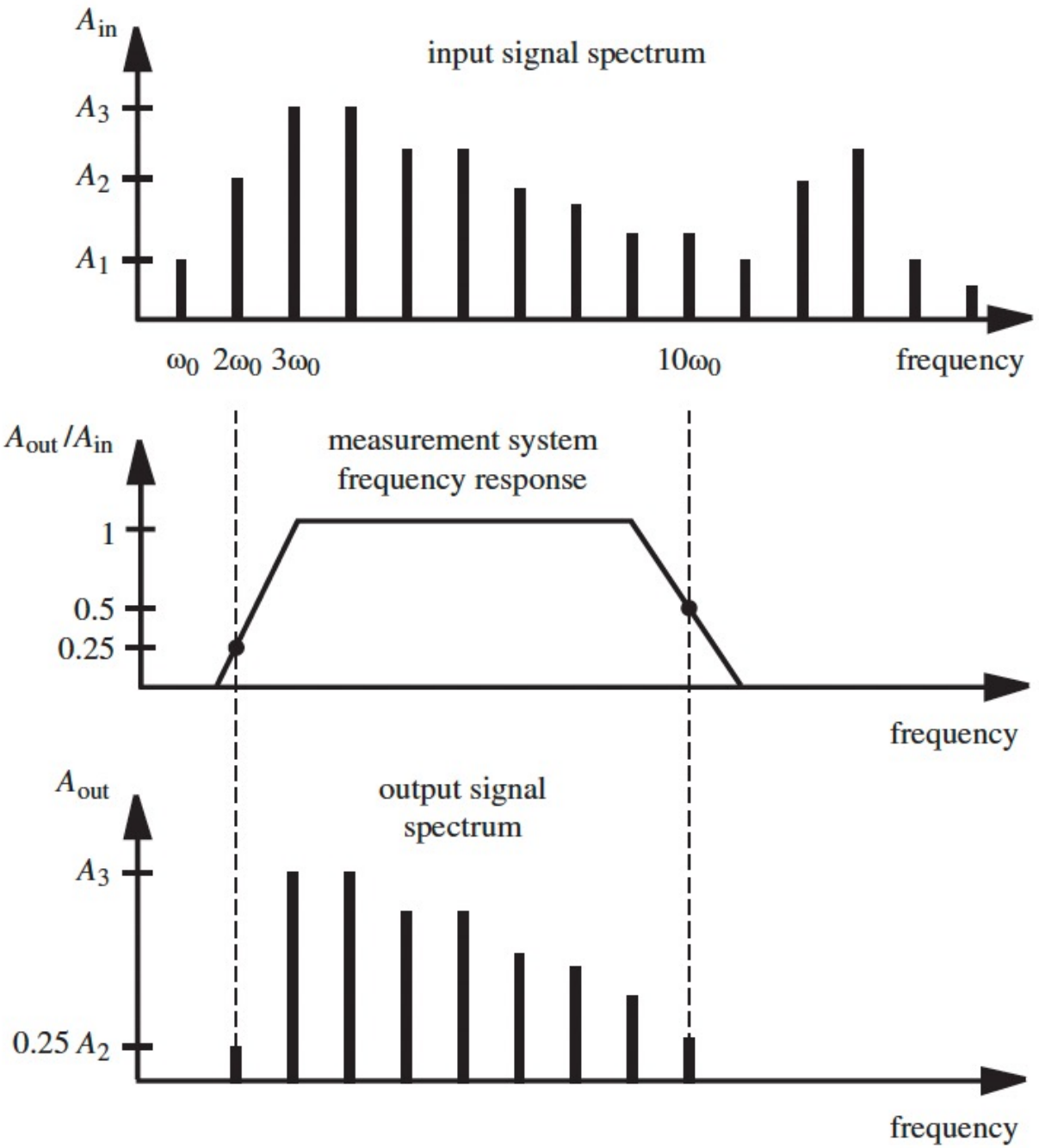


ADC output
Sampled at discrete time intervals
Quantized to the nearest representable discrete value of the amplitude



Fully digitized data (ADC output)

Effect of measurement system bandwidth on a signal



Reference

- Alexander, Charles K., Matthew NO Sadiku, and Matthew Sadiku. Fundamentals of electric circuits. Boston: McGraw-Hill Higher Education.
- Jouaneh, Musa. Fundamentals of mechatronics. Cengage Learning.
- Patrick F Dunn , Measurement and Data Analysis for Engineering and Science.
- Ogata, Modern Control Engineering.
- David G. Alciatore, Introduction to Mechatronics and Measurement Systems
- Carryer, J. Edward, R. Matthew Ohline, and Thomas William Kenny. Introduction to mechatronic design